TEAM #5

Decoding El Niño: Data Driven Analysis and Prediction

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Agenda

01 What is El Niño?

Conquering the climate phenomenon.

02 Objective

Predicting El Niño for given months in advance.

03 Research

Identifying El Niño.

04 Regression Model

Binary classifications and characteristics.

05 Regressing Linearly

Refurbishing our model to be more honest.

06 Conclusions

Takeaways

01

What is El Niño?

What is El Niño 🤔

Classified

by the warming of ocean surface temperatures in the central and eastern tropical Pacific Ocean.

Weather

can be observed as dramatic seasonal changes including shifted rainfall, floods, and droughts.

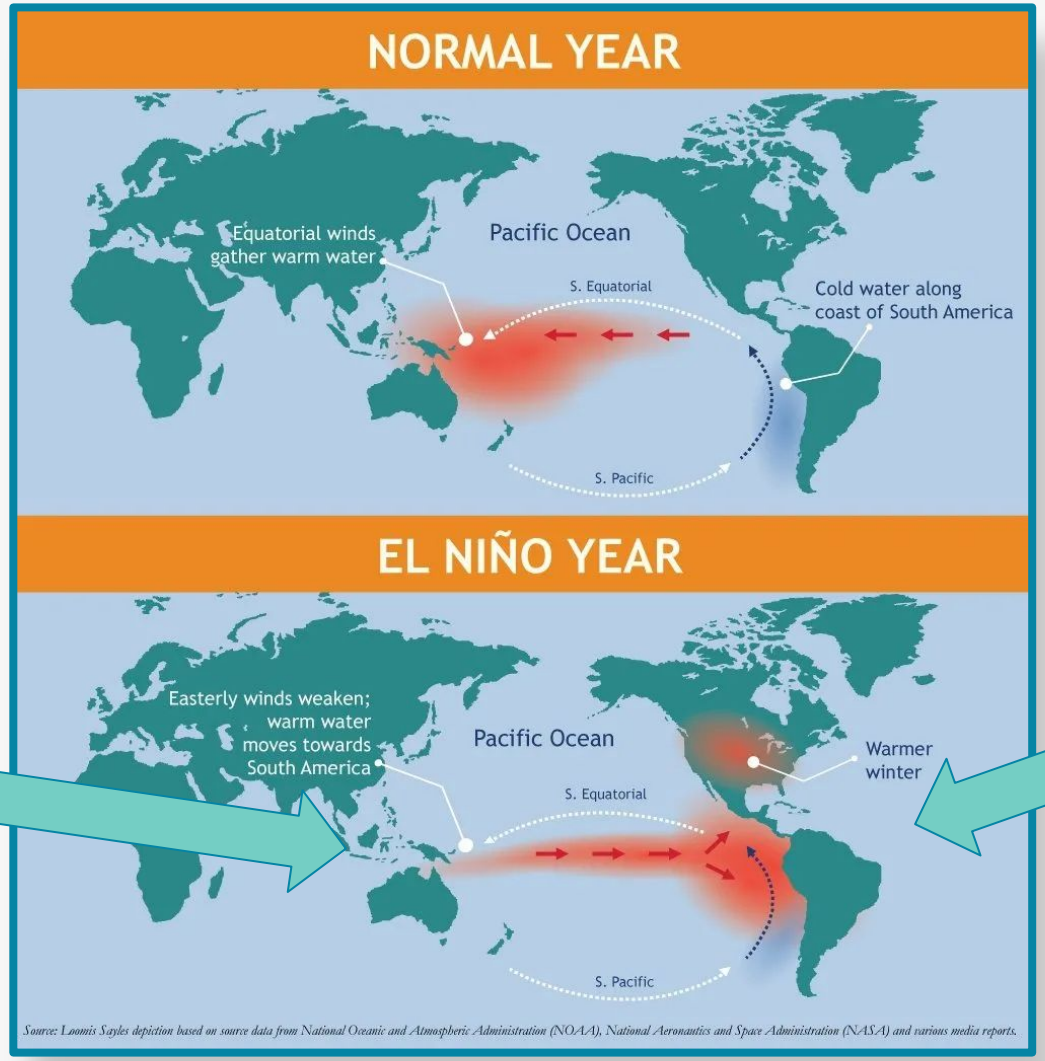
Event

every 3–7 years and lasting 9–12 months, exclusively in the central and eastern equatorial Pacific Ocean, yet affecting much more.

Problems

manifest as agricultural losses, marine industry loss, damage to infrastructure, public health risks.

Significant reduced rainfall over Indonesia.

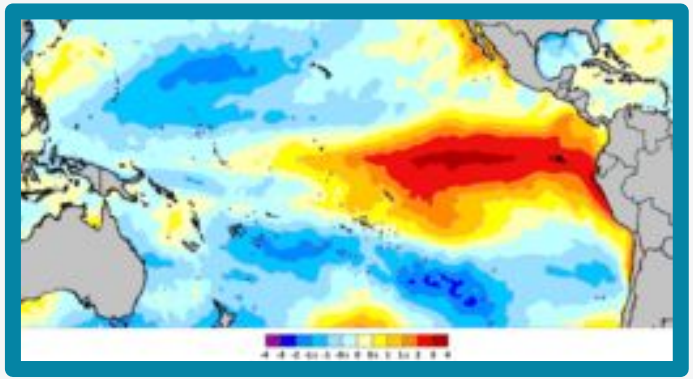


Increased rainfall in the eastern Pacific.

1982–1983 ENSO event

Strongest El Nino event of the **century**

Led to **droughts** in **Indonesia** and **Australia**
Widespread **flooding** across **southern United States**
Abnormal amount of **hurricanes** in the **Pacific Ocean**



~4.1 trillion USD in damages

Inability to **predict** and **prepare** for this event
highlighted the need for a **predictive modeling system**.

El Niño impacts

~60,000,000

people through droughts,
floods, and hurricanes

02

Objective

Project Objectives

Analysis

Analyze historical climate data associated with El Niño events.

Identification

Identify key patterns and indicators linked to El Niño development.

Application

Apply data analysis and modeling techniques to decode El Niño behavior.

Developmental Evaluation

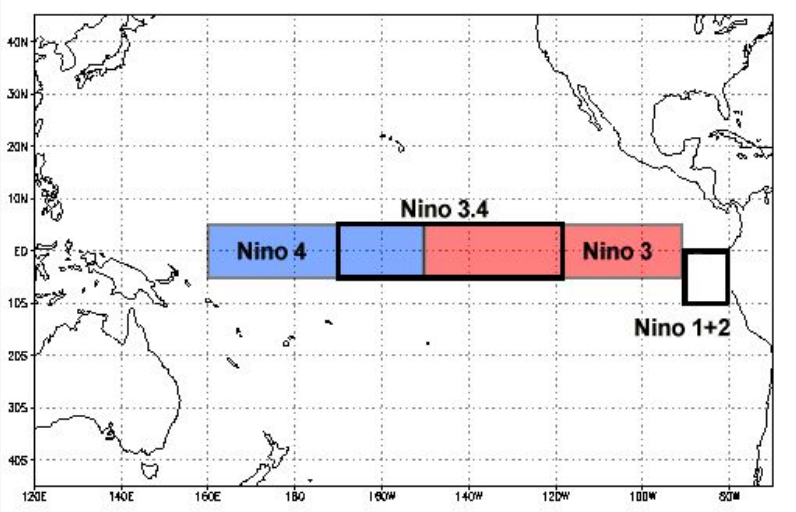
Develop a predictive model to forecast El Niño conditions. Later evaluate model performance using appropriate metrics.

03

Research

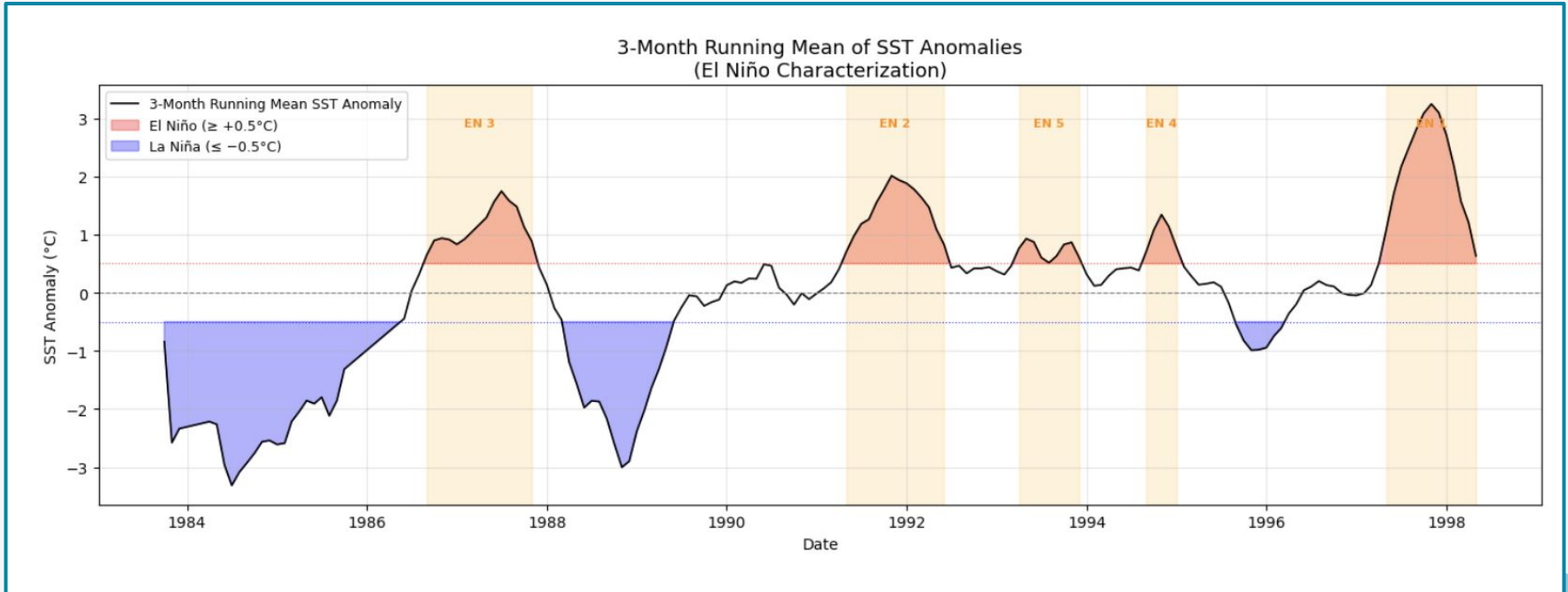
El Nino Numeric Definition

"**El Niño** (La Niña) is a phenomenon in the **equatorial Pacific Ocean** characterized by a **five consecutive 3-month running mean** of **sea surface temperature (SST) anomalies** in the **Niño 3.4 region** that is **above** (below) the **threshold** of **+0.5ºC** (-0.5ºC). This standard of measure is known as the **Oceanic Niño Index (ONI)**."



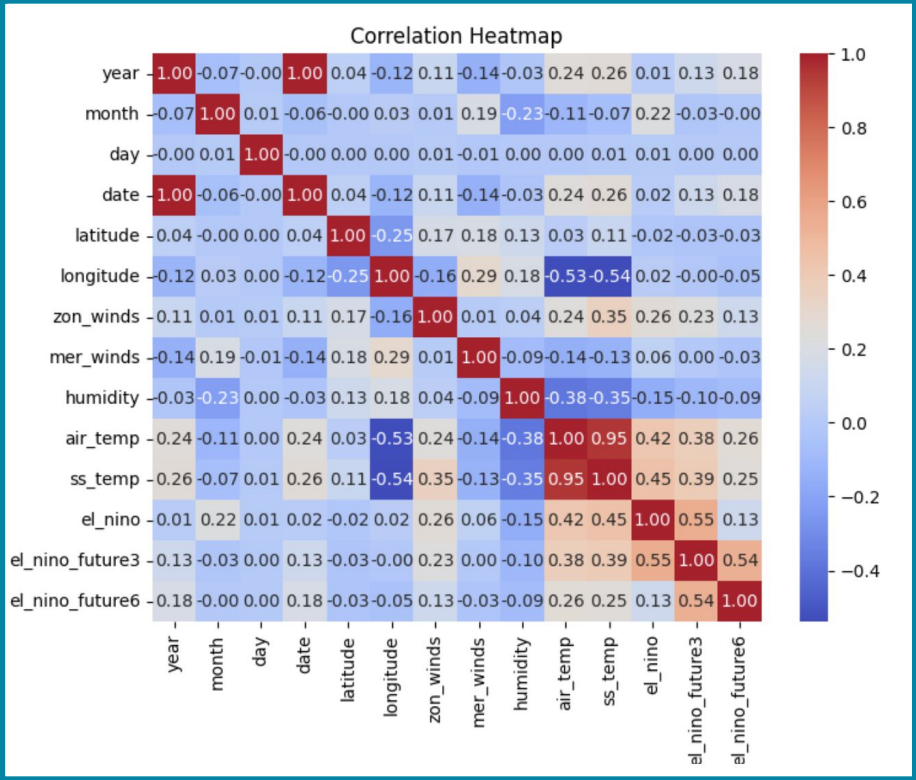
Source: https://tinyurl.com/ja9sps6w

3-Month Rolling Average of SST Anomalies

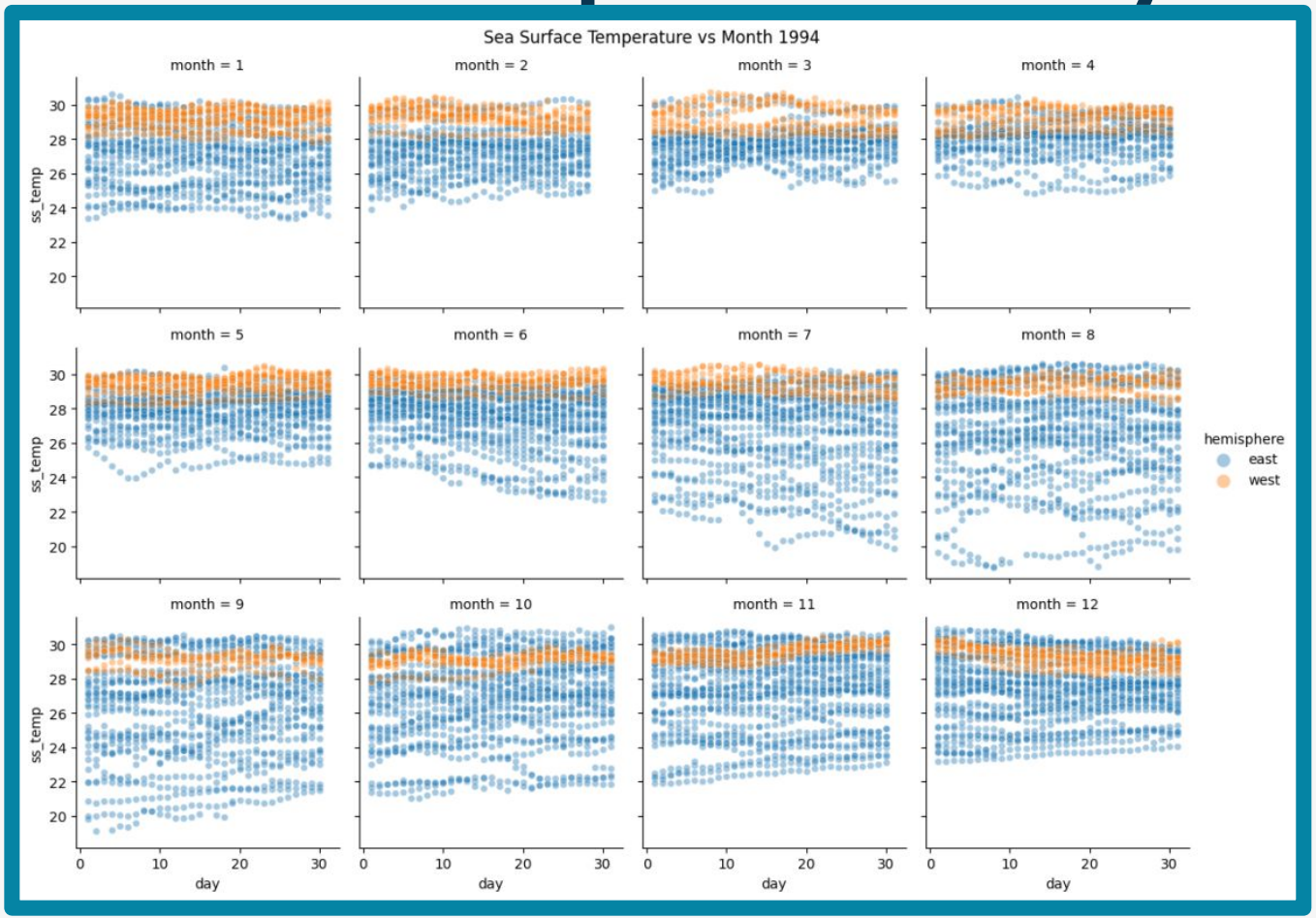


Identifying El Niño

- Correlation between air temperatures and sea surface temperatures
- Moderate correlation between El Niño in 3 months in advance and El Niño in 6 months in advance



Sea Surface Temperature Analysis



04

Regression Model

Is it currently an El Niño?

Splitting data

- Ordinal data
- Split on a date
- ~70% to training

Logistic Regression

```
features = [
    "zon_winds",
    "mer_winds",
    "humidity",
    "air_temp",
    "latitude",
    "longitude",
    "month"
]

target = "el_nino"
```

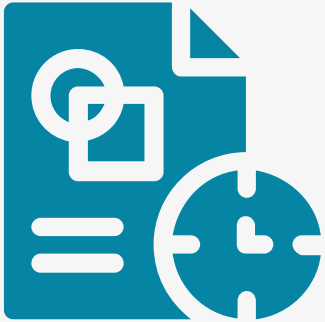
Results

- AUC: ~0.923
- This makes sense

Three month prediction?

Shifting data

- Shifted target 3 months forward
- Similar split



Logistic Regression

```
features = [
    "zon_winds",
    "mer_winds",
    "humidity",
    "air_temp",
    "latitude",
    "longitude",
    "month"
]

target = el_nino_future3
```

Results

- AUC: ~0.866

05

Regressing

Three month prediction?

Features

```
features = [
    "zon_winds",
    "mer_winds",
    "humidity",
    "air_temp",
    "latitude",
    "longitude",
    "month"
]

target = el_nino_future3
```

vs. Baseline

Current status

```
features = [
    "el_nino_now”
]

target = el_nino_future3
```

AUC: ~0.866

AUC Difference: ~0.02

AUC: ~0.846

Six month prediction?

Features

```
features = [
    "zon_winds",
    "mer_winds",
    "humidity",
    "air_temp",
    "latitude",
    "longitude",
    "month"
]

target = el_nino_future6
```

vs. Baseline

Current status

```
features = [
    "el_nino_now"
]

target = el_nino_future6
```

AUC: ~0.678

AUC Difference: ~0.03

AUC: ~0.708

Six month prediction?

Features

```
features = [
    "zon_winds",
    "mer_winds",
    "humidity",
    "air_temp",
    "latitude",
    "longitude",
    "month"
]

target = el_nino_future6
```

vs. Baseline

Due to Niño inertia

Current status

```
features = [
    "el_nino_now"
]

target = el_nino_future6
```

AUC: ~0.678

AUC Difference: ~0.03

AUC: ~0.708

New strategy: Aggregating months

Originally

- Many rows per month:
  - Latitude
  - Longitude
  - Date

Now

- One row per month
  - Averaged features into one month row

New 6 month prediction

Using one row per month:

Logistic AUC: **0.891**

Persistence AUC: **0.727**

Logistic beats persistence by ~**0.17** (big) :)

R^2=0.053

So now only **5.3%** of variance in the prediction is explained by current state of El Niño.

Less than half the R^2 from before aggregating months

06

Conclusions

To conclude...

Ocean-Atmosphere Link

Research validated strong correlation between sea surface temperatures and atmospheric conditions, fundamental driver of El Niño phenomenon.

Effective Predictive Modeling

Successfully utilized binary classification for forecasting. Model predicts El Niño with 3-6 month anticipation. (AUC of ~0.866, AUC of ~0.708).

Climate Variability

Longer time horizons introduce prediction volatility, current conditions are powerful predictors. More climate variable needed for a more robust forecast.

To conclude...

3 Month vs. 6 month

At 3 months, predictions are heavily inertia-based..

- Current El Niño state persists forward with minimal atmosphere interference
- Strong predictive performance allowing current conditions to serve as reliable proxies for near-future states

At 6 months, predictions are heavily susceptible

- Increasing influence of stochastic atmospheric variability beyond seasonal inertia window
- Dependent on precursor signals from winds, humidity, and temperature patterns

Thank you Azra for being a wonderful mentor!

Thanks!